

## Topic P3: Electricity

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### P3.1 Static and charge

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#### Summary

Having established the nature of matter, consideration is now given to the interactions between matter and electrostatic fields. These interactions are derived from the structure of matter which was considered in Topic P1. The generation of charge is considered. Charge is a fundamental property of matter. There are two types of charge which are given the names 'positive' and 'negative'.

#### Underlying knowledge and understanding

Learners should be aware of electron transfer leading to objects becoming statically charged and the forces between them. They should also be aware of the existence of an electric field.

#### Common misconceptions

Learners commonly have difficulty classifying materials as insulators or conductors. They find it difficult to remember that positive charge does not move to make a material positive, rather it is the movement of electrons.

#### Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

| Reference | Mathematical learning outcomes                             | Mathematical skills          |
|-----------|------------------------------------------------------------|------------------------------|
| PM3.1i    | recall and apply: charge flow (C) = current (A) × time (s) | M1a, M2a, M3a, M3b, M3c, M3d |

| Topic content                             |                                                                                                                                                                                                | Opportunities to cover:      |                                      | Practical suggestions                                                                                                                                                                 |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learning outcomes                         | To include                                                                                                                                                                                     | Maths                        | Working scientifically               |                                                                                                                                                                                       |
| P3.1a                                     | describe that charge is a property of all matter and that there are positive and negative charges                                                                                              |                              | WS1.1b, WS1.1e, WS1.2a, WS1.3e, WS2a | Use of charged rods to repel or attract one another.<br><br>Use of a charged rod to deflect water or pick up paper.<br><br>Discussion of why charged balloons are attracted to walls. |
| P3.1b                                     | describe the production of static electricity, and sparking, by rubbing surfaces, and evidence that charged objects exert forces of attraction or repulsion on one another when not in contact |                              | WS1.1b, WS1.1e, WS1.2a, WS1.3e       | Use of a Van de Graaff generator.                                                                                                                                                     |
| P3.1c                                     | explain how transfer of electrons between objects can explain the phenomena of static electricity                                                                                              |                              | WS1.1b, WS1.3e, WS1.3f, WS2a         | Use of the gold leaf electroscope and a charged rod to observe and discuss behaviour.                                                                                                 |
| P3.1d <input checked="" type="checkbox"/> | explain the concept of an electric field and how it helps to explain the phenomena of static electricity                                                                                       | M5b                          | WS1.3e                               | Demonstration of semolina on castor oil to show electric fields.                                                                                                                      |
| P3.1e                                     | recall that current is a rate of flow of charge (electrons) and the conditions needed for charge to flow                                                                                       |                              |                                      |                                                                                                                                                                                       |
| P3.1f                                     | recall that current has the same value at any point in a single closed loop                                                                                                                    |                              |                                      |                                                                                                                                                                                       |
| P3.1g                                     | recall and use the relationship between quantity of charge, current and time                                                                                                                   | M1a, M2a, M3a, M3b, M3c, M3d |                                      |                                                                                                                                                                                       |